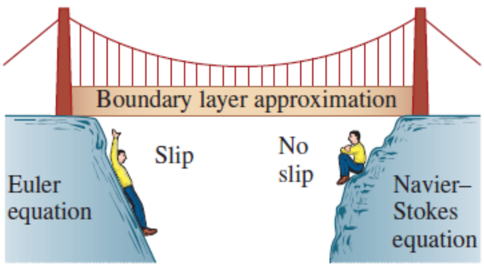
Boundary Layer:

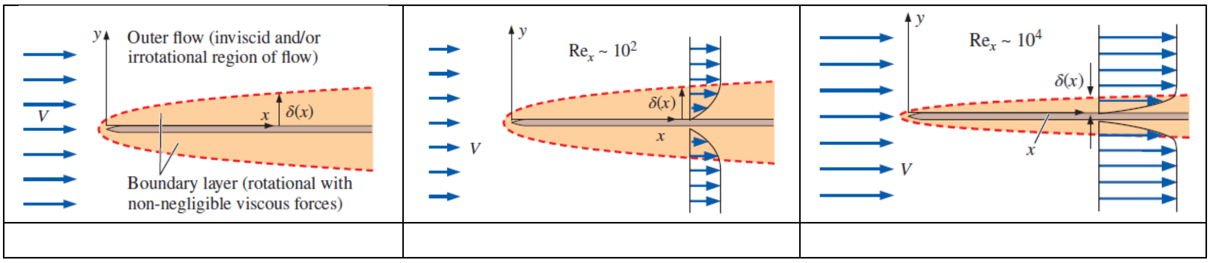
From a historical perspective, by the mid-1800s, the Navier–Stokes equation was known, but could not be solved except for flows of very simple geometries.

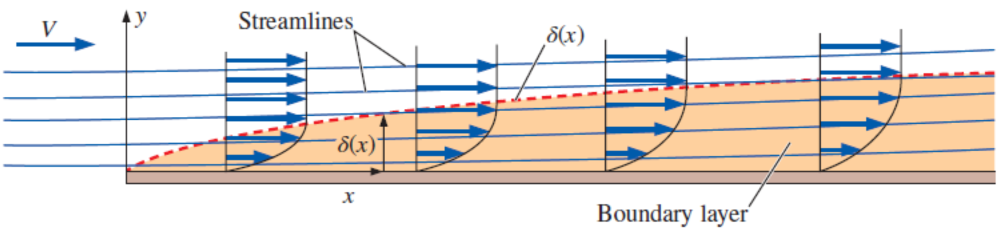
A major breakthrough in fluid mechanics occurred in 1904 when Ludwig Prandtl (1875–1953) introduced the **boundary layer approximation**. Prandtl's idea was to divide the flow into two regions: an **outer flow region** that is inviscid and/or irrotational, and an inner flow region called a **boundary layer.**

The boundary layer approximation bridges the gap between the Euler equation and the Navier–Stokes equation, and between the slip condition and the no-slip condition at solid walls.

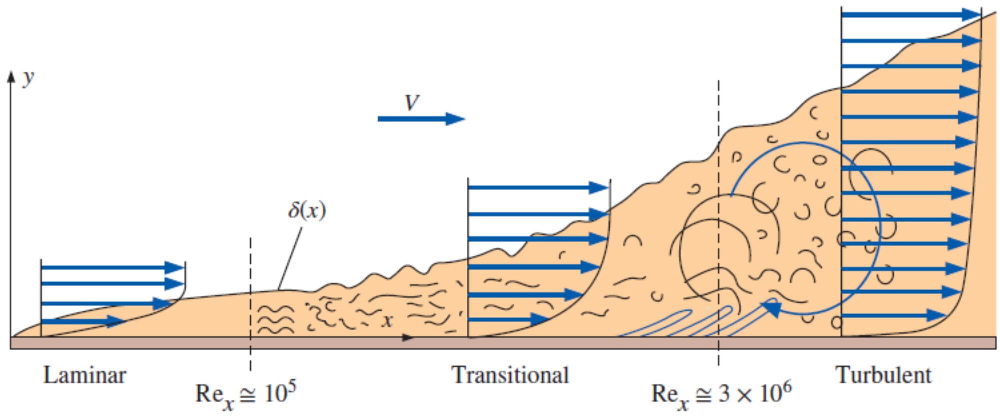


Boundary layer - A very thin region of flow near a solid wall where viscous forces and rotationality cannot be ignored.

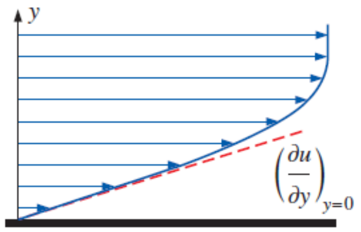




Comparison of streamlines and the curve representing δ as a function of x for a flat plate boundary layer. Since streamlines cross the curve $\delta(x)$, $\delta(x)$ cannot itself be a streamline of the flow.



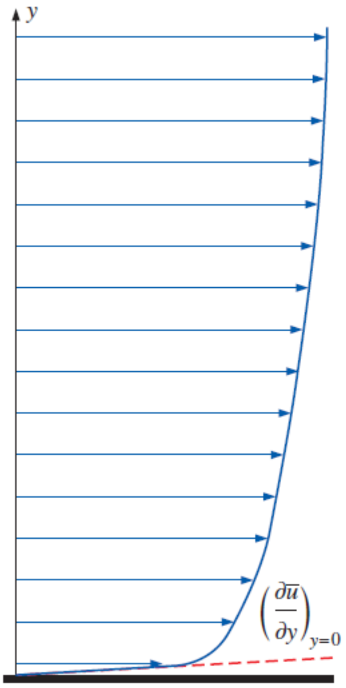
The velocity gradients at the wall, and thus the wall shear stress, are much larger for turbulent flow than they are for laminar flow, even though the turbulent boundary layer is thicker than the laminar one for the same value of free-stream velocity.



Laminar flow

$$\boldsymbol{\tau_{laminar} \ll \tau_{turbulent}}$$

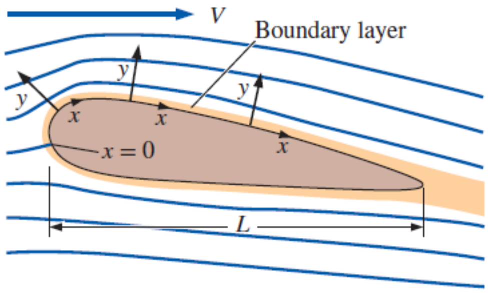
$$\boldsymbol{\frac{\partial u}{\partial y} \rightarrow \text{steeper for turbulent flow}}$$



Turbulent flow

Laminar Boundary Layer

- Consider a flat plate moving in a fluid or a fluid passing by a solid plate.
- The boundary layer coordinate system for flow over a body; *x* follows the Surface and is typically set to zero at the front stagnation point of the body, and *y* is everywhere normal to the surface locally.



- Boundary layer grows downstream (**gradient along x-direction cannot be ignored**).
- Boundary layer is a thin layer important for aerodynamics, heat transfer, and atmospheric flow.
- Near the wall – no-slip boundary condition (means velocity close to the wall has to have the velocity of the wall itself due to friction)

Assumptions:

- Incompressible (ρ = constant)
- 2D (Dimensional)/ 2C (Component) (w = 0; $\partial/\partial z = 0$)
- No body forces

The boundary layer grows in the x-direction. X-gradient in the velocity field cannot be ignored.

Boundary conditions to solve the equations:

At the wall, we have no-slip and no-penetration conditions.

$\mathbf{No-slip}: \mathrm{u, w = 0\ at\ y = 0}$

$\mathbf{No-penetration}: \mathrm{v = 0\ at\ y = 0}$

The incompressible continuity equation in two dimensions,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

The incompressible momentum equations in two dimensions,

$$\mathbf{x-direction} \rightarrow \rho\left[\cancel{\frac{\partial \mathbf{u}}{\partial t}} + u\frac{\partial \mathbf{u}}{\partial x} + v\frac{\partial \mathbf{u}}{\partial y} + \cancel{w\frac{\partial \mathbf{u}}{\partial z}}\right] = -\frac{\partial P}{\partial \mathbf{x}} + \mu\left(\frac{\partial^2 \mathbf{u}}{\partial x^2} + \frac{\partial^2 \mathbf{u}}{\partial y^2} + \cancel{\frac{\partial^2 \mathbf{u}}{\partial z^2}}\right) + \cancel{\rho a_{\mathbf{x}}}$$

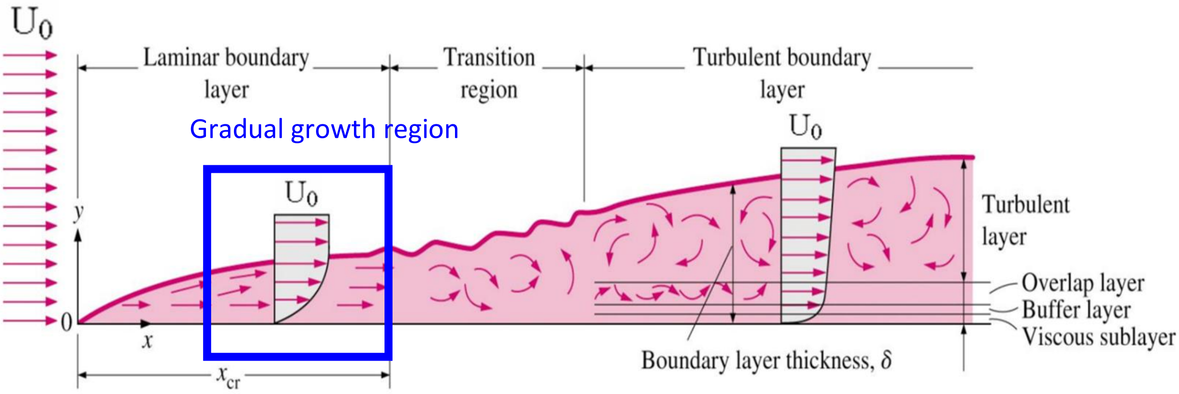
$$\mathbf{y-direction} \rightarrow \rho\left[\cancel{\frac{\partial \mathbf{v}}{\partial t}} + u\frac{\partial \mathbf{v}}{\partial x} + v\frac{\partial \mathbf{v}}{\partial y} + \cancel{w\frac{\partial \mathbf{v}}{\partial z}}\right] = -\frac{\partial P}{\partial \mathbf{y}} + \mu\left(\frac{\partial^2 \mathbf{v}}{\partial x^2} + \frac{\partial^2 \mathbf{v}}{\partial y^2} + \cancel{\frac{\partial^2 \mathbf{v}}{\partial z^2}}\right) + \cancel{\rho a_{\boldsymbol{y}}}$$

$$\rho\left[u\frac{\partial \mathbf{u}}{\partial x} + v\frac{\partial \mathbf{u}}{\partial y}\right] = -\frac{\partial P}{\partial \mathbf{x}} + \mu\left(\frac{\partial^2 \mathbf{u}}{\partial x^2} + \frac{\partial^2 \mathbf{u}}{\partial y^2}\right)$$

$$\rho\left[u\frac{\partial \mathbf{v}}{\partial x} + v\frac{\partial \mathbf{v}}{\partial y}\right] = -\frac{\partial P}{\partial \mathbf{y}} + \mu\left(\frac{\partial^2 \mathbf{v}}{\partial x^2} + \frac{\partial^2 \mathbf{v}}{\partial y^2}\right)$$

Pressure may change *along* a boundary layer (*x*-direction), but the change in pressure *across* a boundary layer (*y*-direction) is negligible.

By considering only the **gradual growth region**, we can make some more possible assumptions.



Assumptions:

- Growth is slow in x-direction. Hence,
- $\boldsymbol{v \ll u}$ *and* $\boldsymbol{\partial/\partial x \ll \partial/\partial y}$

$$\frac{\partial u}{\partial y} \gg \frac{\partial u}{\partial x} \text{ and } \frac{\partial^2 u}{\partial y^2} \gg \frac{\partial^2 u}{\partial x^2}$$

$$\frac{\partial v}{\partial y} \gg \frac{\partial v}{\partial x} \text{ and } \frac{\partial^2 v}{\partial y^2} \gg \frac{\partial^2 v}{\partial x^2}$$

Terms that causes **significant** impact:

$$\mathbf{u, \partial u/\partial y\,, \partial^2 u/\partial y^2}$$

Terms that causes **small** impact:

$$\mathbf{v, \partial u/\partial x\,, \partial^2 u/\partial x^2\,, \partial v/\partial y}$$

Terms that causes **very small** impact:

$$\mathbf{\partial v/\partial x\,, \partial^2 v/\partial x^2}$$

$$\rho\left[\mathbf{u}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} + \mathbf{v}\frac{\partial \mathbf{u}}{\partial \mathbf{y}}\right] = -\frac{\partial P}{\partial \mathbf{x}} + \mu\left(\frac{\partial^2 \mathbf{u}}{\partial \mathbf{x}^2} + \frac{\partial^2 \mathbf{u}}{\partial \mathbf{y}^2}\right)$$

Keep the significant, significant times small, and significant times very small terms in the two equations. Ignore the other terms.

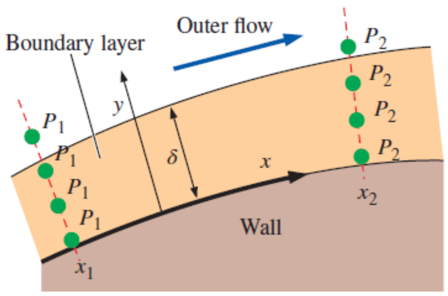
$$\rho\left[\mathbf{u}\frac{\partial \mathbf{v}}{\partial \mathbf{x}} + \mathbf{v}\frac{\partial \mathbf{v}}{\partial \mathbf{y}}\right] = -\frac{\partial P}{\partial \mathbf{y}} + \mu\left(\frac{\partial^2 \mathbf{v}}{\partial \mathbf{x}^2} + \frac{\partial^2 \mathbf{v}}{\partial \mathbf{y}^2}\right)$$

Comparing the above two equations, we can see that all terms in the first equation are much bigger than their partner terms in second equation. All terms in the first equation are greater than the second terms. Also,

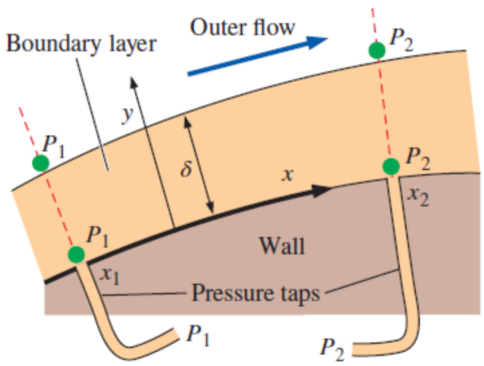
$$\frac{\partial P}{\partial x} \gg \frac{\partial P}{\partial y}$$

Pressure is not a function of y and it is approximately only a function of x.

Pressure may change *along* a boundary layer (*x*-direction), but the change in pressure *across* a boundary layer (*y*-direction) is negligible. Physically, because the boundary layer is so thin, streamlines within the boundary layer have negligible *curvature* when observed at the scale of the boundary layer thickness.



Since the streamlines are not significantly curved in a thin boundary layer, there is no significant pressure gradient across the boundary layer.
The pressure in the irrotational region of flow outside of a boundary layer can be measured by static pressure taps in the surface of the wall.



From the boundary condition, we know that,

$$\left.\frac{\partial \mathbf{P}}{\partial \mathbf{x}}\right|_{y\to\infty} = \mathbf{0}$$

This means that pressure gradient is zero everywhere because it is constant in y-direction.

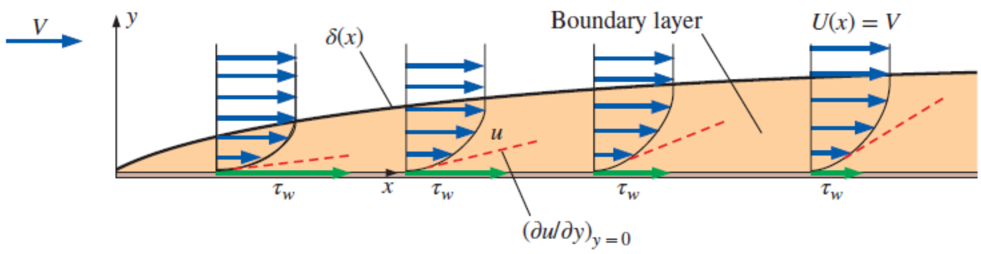
By an order-of-magnitude analysis, the complete Navier-Stokes equations for two-dimensional flow reduce to the following boundary-layer equations:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow \textbf{Continuity equation}$$

$$\rho\left[u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right] = \mu\left(\frac{\partial^2 u}{\partial y^2}\right) \rightarrow \textbf{Momentum equation along x}$$

$$\frac{\partial P}{\partial y} = 0 \rightarrow \textbf{Momentum equation along y}$$

These equations are still complicated and solvable **numerically**. In 1908 Prandtl and his student Blasius derived these equations and solved it. The solution of these equations is called **Blasius solution**.



For a laminar flat plate boundary layer, wall shear stress decays like x−1/2 as the slope $\partial u/\partial y$ at the wall decreases downstream. The front portion of the plate contributes more skin friction drag than does the rear portion.

To analyse Boundary Layer, we need

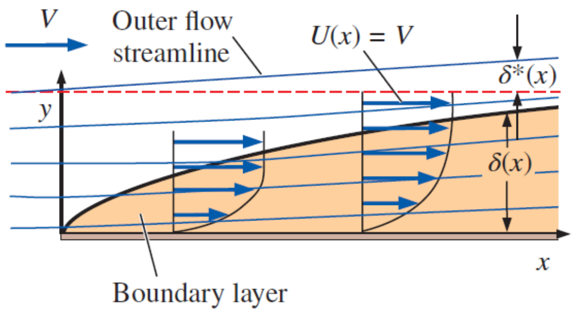
$$\mathbf{Re} = \frac{\rho u x}{\mu}$$

where x = length scale (local distance from the leading edge); μ = dynamic viscosity of the fluid.

For laminar flows, we have the numerical solution and we can define the behaviour exactly through **Disturbance thickness (δ)**, **Displacement thickness (δ^*)**, and **Momentum thickness (θ)** as they vary in the streamwise direction.

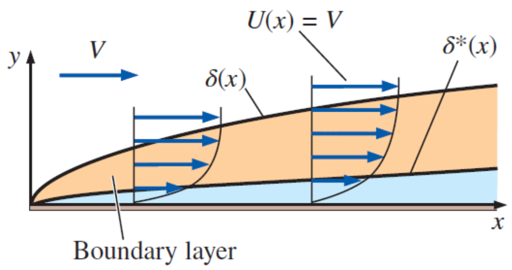
Displacement Thickness:

Displacement thickness is the distance that a streamline just outside of the boundary layer is deflected away from the wall due to the effect of the boundary layer.

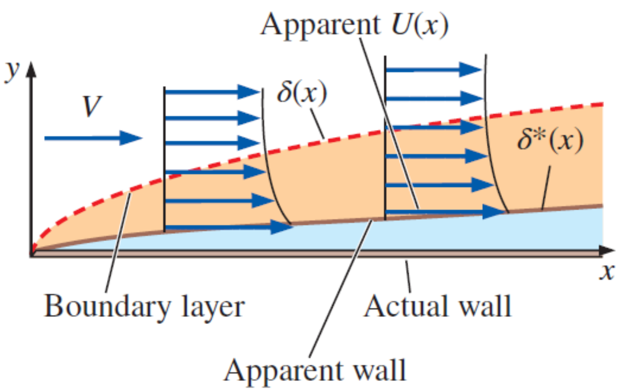


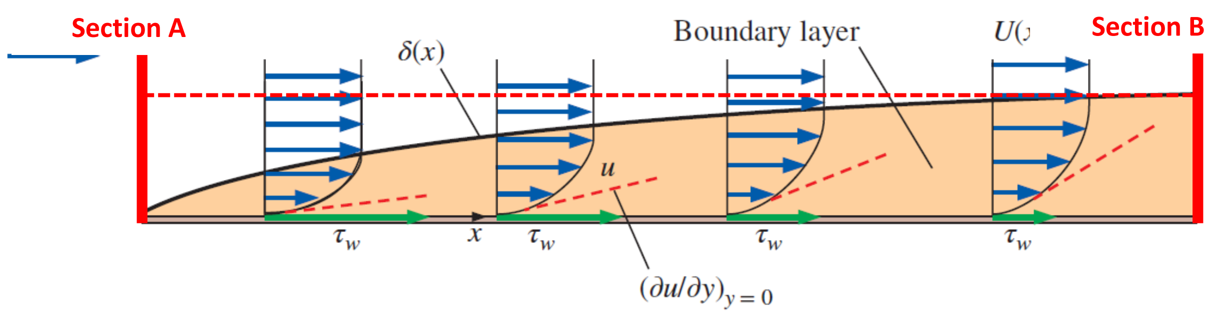
Displacement thickness is the imaginary increase in thickness of the wall, as seen by the outer flow, due to the effect of the growing boundary layer.

For a laminar flat plate boundary layer, the displacement thickness is roughly one-third of the 99 percent boundary layer thickness.



The boundary layer affects the irrotational outer flow in such a way that the wall appears to take the shape of the displacement thickness. The apparent U(x) differs from the original approximation because of the “thicker” wall.





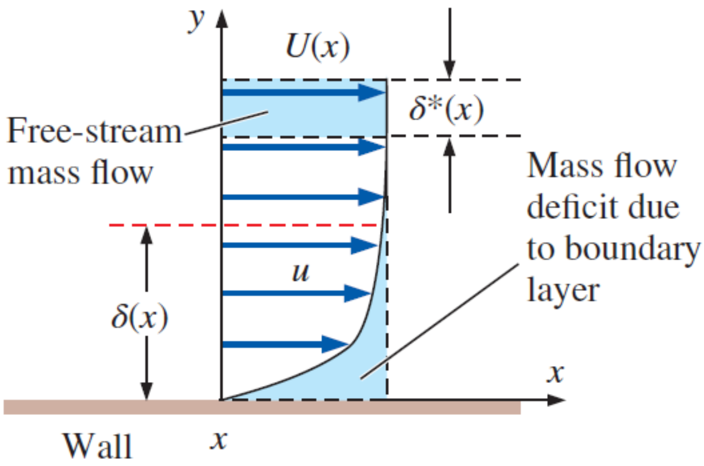
Mass flow rate through section A,

$$\dot{\mathbf{m}}_{\mathbf{A}} = \boldsymbol{\rho}\mathbf{U}_{\infty}\boldsymbol{\delta} = \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{U}_{\infty}\mathbf{dy}$$

Mass flow rate through section B,

$$\dot{\mathbf{m}}_{\mathbf{B}} = \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{u}(y)dy + \int_{\delta}^{\delta+\delta^*} \boldsymbol{\rho}\mathbf{U}_{\infty}\mathbf{dy}$$

Physically, the mass flow *deficit* within the boundary layer (the lower blue shaded region) is replaced by a chunk of free-stream flow of thickness δ^* (the upper blue-shaded region).



Comparison of the area under the boundary layer profile, representing the mass flow deficit, and the area generated by a chunk of free-stream fluid of thickness δ^*. To satisfy conservation of mass, these two areas must be identical.

From the conservation of mass,

$$\dot{\mathbf{m}}_{\mathbf{A}} = \dot{\mathbf{m}}_{\mathbf{B}} \Rightarrow \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{U}_{\infty}\mathbf{dy} = \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{u}(\mathbf{y})\mathbf{dy} + \int_{\delta}^{\delta+\delta^*} \boldsymbol{\rho}\mathbf{U}_{\infty}\mathbf{dy}$$

$$\int_{0}^{\delta} \boldsymbol{\rho}\mathbf{U}_{\infty}\mathbf{dy} = \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{u}(\mathbf{y})\mathbf{dy} + \boldsymbol{\rho}\mathbf{U}_{\infty}\boldsymbol{\delta}^* \Rightarrow \boldsymbol{\rho}\mathbf{U}_{\infty}\boldsymbol{\delta}^* = \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{U}_{\infty}\mathbf{dy} - \int_{0}^{\delta} \boldsymbol{\rho}\mathbf{u}(\mathbf{y})\mathbf{dy}$$

$$\boldsymbol{\delta}^* = \int_{0}^{\delta} \left(1 - \frac{\mathbf{u}(\mathbf{y})}{\mathbf{U}_{\infty}}\right)\mathbf{dy}$$

Momentum Thickness:

Another measure of boundary layer thickness is **momentum thickness**, commonly given the symbol θ.

Since the bottom of the control volume is the plate itself, no mass or momentum can cross that surface. The top of the control volume is taken as a streamline of the outer flow. Since no flow can cross a streamline, there can be no mass or momentum flux across the upper surface of the control volume.

From the conservation of momentum (p),

$$\mathbf{p_A = p_B \Rightarrow \int_0^{\delta} (\rho U_\infty dy)\, U_\infty = \int_0^{\delta} (\rho u(y) dy) u(y) + \int_{\delta}^{\delta+\theta} (\rho U_\infty dy) U_\infty}$$

$$\mathbf{\int_0^{\delta} (\rho U_\infty dy)\, U_\infty = \int_0^{\delta} (\rho u(y) dy) u(y) + \rho U_\infty^2 \theta \Rightarrow \rho U_\infty^2 \theta = \int_0^{\delta} (\rho U_\infty dy)\, U_\infty - \int_0^{\delta} (\rho u(y) dy) u(y)}$$

$$\mathbf{\theta = \int_0^{\delta} \frac{u(y)}{U_\infty} \left(1 - \frac{u(y)}{U_\infty}\right) dy}$$

Momentum thickness is defined as the loss of momentum flux per unit width divided by ρU^2 due to the presence of the growing boundary layer.

Property	Laminar
Boundary layer thickness	$\frac{\delta}{x} = \frac{4.91}{\sqrt{Re_x}}$
Displacement thickness	$\frac{\delta^*}{x} = \frac{1.72}{\sqrt{Re_x}}$
Momentum thickness	$\frac{\theta}{x} = \frac{0.664}{\sqrt{Re_x}}$
Local skin friction coefficient	$C_{f,x} = \frac{0.664}{\sqrt{Re_x}}$

Disturbance thickness (δ), **Displacement thickness (** δ^* **)**, and **Momentum thickness (θ) are the functions of local Reynolds number (Re_x).**

The laminar boundary layer grows downstream non-linearly because x appears in the square root of the Reynolds number.

Boundary layer create wall shear. From **Newton's law of viscous shear**,

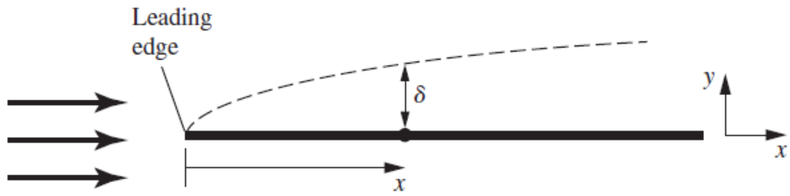
$$\mathbf{\tau_w = \mu \left(\frac{\partial u}{\partial y}\right)_{y=0}}$$

Numerically,

$$\tau_{wall} = 0.332\mu(U_\infty/x)\sqrt{Re_x}$$

Further down (x) the stream, **shear stress will be less**.

As the **boundary layer grows**, the **velocity gradient** near the wall gets **smaller**.



Drag force:

$$F = \int_A \tau_w dA = \int_0^L 0.332\mu(U_\infty/x)\sqrt{Re_x}(W)(dx)$$

$$F = W\int_0^L 0.332\mu(U_\infty/x)\left(\frac{\rho U_\infty x}{\mu}\right)^{-1/2} dx$$

$$F = 0.664\rho U_\infty^2 L(Re_L)^{-1/2}$$

Turbulent Boundary Layer:

- In reality, most flows are turbulent.
- On an aircraft, flow becomes turbulent within 24 mm of travelling on the aircraft. For ship – 32 mm and For Tuna (fish) – 50 mm.
- At high Reynolds number, Boundary layer transition from Laminar (clean, calm, organized) to Turbulent (chaotic).
- When flow inertia exceeds its ability to respond/recover to disturbances (viscous force) it transitions to turbulent.
- Turbulent flow is chaotic and unsteady.
- But in the averaged state, it is well behaved and relatively predictable.

Transition: defined by the flow Reynolds number based on the freestream velocity and the distance downstream from where the boundary layer started. This is called Critical Reynolds number.

Critical Reynolds number: At which the flow transitions to turbulence.

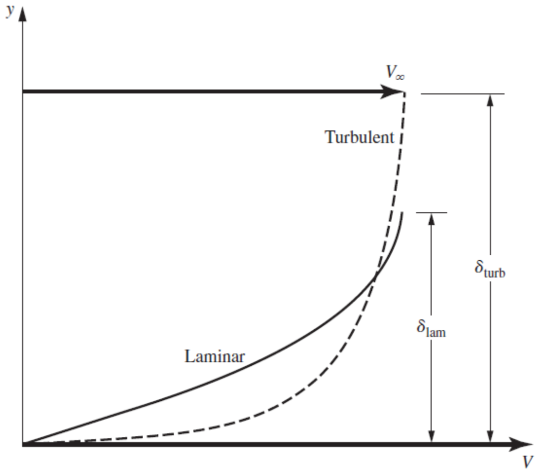
For a smooth flat plate, the critical Re is roughly 5 x 10^5

$$\mathbf{Re_{crit}} \approx \mathbf{5 \times 10^5}$$

$\mathbf{Re_{crit}}$ **depends on surface quality and incoming flow quality.**

- Consider a uniform flow approaching a smooth flat plate, viscous force jumps into action and start to grow a laminar boundary layer.
- As the flow moves downstream, this region of velocity deficit gets bigger.
- Initially, the growth is quite rapid and then followed by a longer region of gradual growth.
- At some point at the critical x location, the flow transition to turbulence.
- At the point of transition, the boundary layer thickness jumps dramatically, then gradually continue its growth.

Laminar and turbulent flows are inherently different and statistically different.



Turbulent flow:

- More energetic than laminar flows.
- Near the wall, there is much **higher velocity**.
- Has **higher shear stress** at the wall (flow pulls on the wall section with a much higher force)
- The **velocity profile** at the wall is **much steeper**. Turbulent mixing brings higher flow velocity from top of the boundary layer closer to the wall.
- Even in time-average state, we do not have full analytical or numerical solutions.
- No analytical solution exists.

- Need to rely on empirical relations. Empirical relations come from collections of measurements and observation.
- Empirical models are limited (**they work well for the cases that were studied and cannot be generally applied**).
- Turbulence quantities are time-averaged (for example, $\bar{u}$).
- One of the models to recreate the time-averaged flow behavior is **Power-law**.

Numerous other empirical velocity profiles exist for turbulent flow. Among those, the simplest and the best known is the **power-law velocity profile** expressed as

$$\frac{\bar{u}}{U_\infty} = \left(\frac{y}{\delta}\right)^n$$

where the exponent *n* is a constant whose value depends on the Reynolds number. The value of *n* increases with increasing Reynolds number. The value *n* = 7 generally approximates many flows in practice, giving rise to the term *one-seventh power-law velocity profile*.

- Power-law works pretty good away from the wall and bad near the wall.

Boundary layer or Disturbance Thickness:

$$\delta = y \text{ where } \bar{u} = 0.99U_\infty$$

Displacement Thickness:

$$\delta^* = \int_0^\delta \left(1 - \frac{\bar{u}(y)}{U_\infty}\right) dy$$

Momentum Thickness:

$$\theta = \int_0^\delta \frac{\bar{u}(y)}{U_\infty}\left(1 - \frac{\bar{u}(y)}{U_\infty}\right) dy$$

Local wall shear stress,

$$\tau_{wall} = \frac{0.0288\rho U_\infty^2}{(Re_x)^{1/5}}$$

Drag force:

$$F = \int_A \tau_w dA = \int_0^L \frac{0.0288\rho U_\infty^2}{(Re_x)^{1/5}}(W)(dx)$$

$$F = W\int_0^L \frac{0.0288\rho U_\infty^2}{(Re_x)^{1/5}} dx = W\frac{0.0288\rho U_\infty^2}{(\rho U_\infty/\mu)^{1/5}}\int_0^L \frac{1}{(x)^{1/5}} dx$$

$$F = \frac{0.036\rho U_\infty^2 WL}{(Re_L)^{1/5}}$$

In non-dimensional form,

$$C_D = \frac{F}{0.5\rho U_\infty^2 WL} = 0.0721(Re_L)^{-1/5}$$

Summary of expressions for laminar and turbulent boundary layers on a smooth flat plate aligned parallel to a uniform stream*

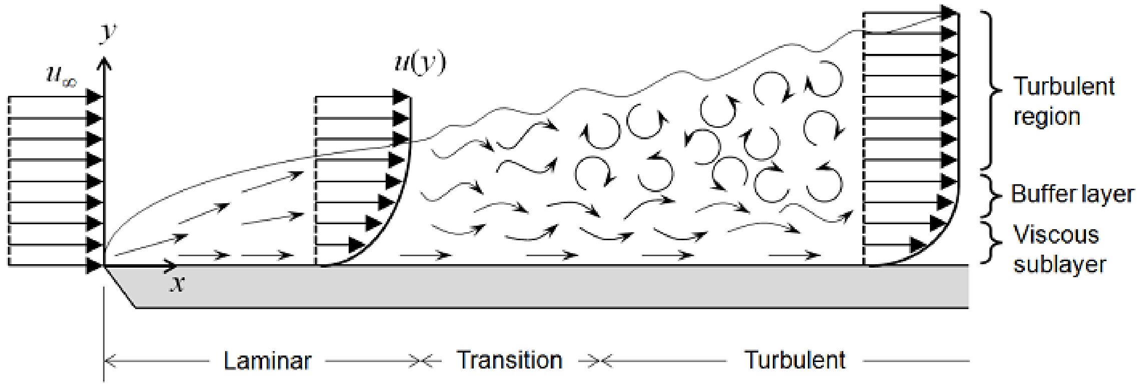
Property	Laminar	(a) Turbulent(†)	(b) Turbulent(‡)
Boundary layer thickness	$\frac{\delta}{x} = \frac{4.91}{\sqrt{Re_x}}$	$\frac{\delta}{x} \cong \frac{0.16}{(Re_x)^{1/7}}$	$\frac{\delta}{x} \cong \frac{0.38}{(Re_x)^{1/5}}$
Displacement thickness	$\frac{\delta^*}{x} = \frac{1.72}{\sqrt{Re_x}}$	$\frac{\delta^*}{x} \cong \frac{0.020}{(Re_x)^{1/7}}$	$\frac{\delta^*}{x} \cong \frac{0.048}{(Re_x)^{1/5}}$
Momentum thickness	$\frac{\theta}{x} = \frac{0.664}{\sqrt{Re_x}}$	$\frac{\theta}{x} \cong \frac{0.016}{(Re_x)^{1/7}}$	$\frac{\theta}{x} \cong \frac{0.037}{(Re_x)^{1/5}}$
Local skin friction coefficient	$C_{f,x} = \frac{0.664}{\sqrt{Re_x}}$	$C_{f,x} \cong \frac{0.027}{(Re_x)^{1/7}}$	$C_{f,x} \cong \frac{0.059}{(Re_x)^{1/5}}$

For turbulent flow from the beginning of the plate,

$$C_D = 0.074(Re_L)^{-1/5} \text{ for } 5\times10^5 < Re_L < 10^7$$

$$C_D = 0.455(\log_{10} Re_L)^{-2.58} \text{ for } 10^7 < Re_L < 10^9$$

Flow with transition (Laminar + Transition + Turbulent):



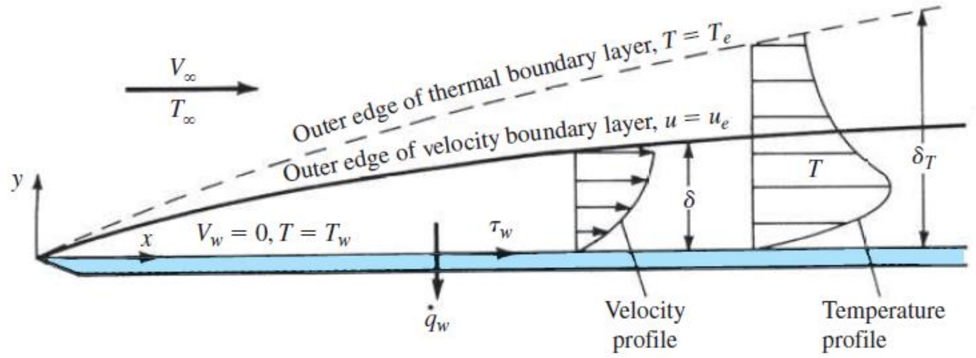
Drag Coefficient:

$$C_D = 0.074(Re_L)^{-1/5} - (1740/Re_L) \textbf{ for } 5 \times 10^5 < Re_L < 10^7$$

$$C_D = 0.455(\log_{10} Re_L)^{-2.58} - (1610/Re_L) \textbf{ for } 5 \times 10^5 < Re_L < 10^9$$

Comparison between laminar and turbulent boundary layer:

1. $(\mathbf{dV/dy})_{\mathbf{y=0}}$ for laminar flow $<$ $(\mathbf{dV/dy})_{\mathbf{y=0}}$ for turbulent flow
2. $(\boldsymbol{\tau}_{\mathbf{w}})_{\mathbf{y=0}}$ for laminar flow $<$ $(\boldsymbol{\tau}_{\mathbf{w}})_{\mathbf{y=0}}$ for turbulent flow
3. $(\mathbf{dT/dy})_{\mathbf{y=0}}$ for laminar flow $<$ $(\mathbf{dT/dy})_{\mathbf{y=0}}$ for turbulent flow



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1. Anderson, J. (2011)*: Fundamentals of Aerodynamics (SI units)*. McGraw hill.
2. Cengel, Y., & Cimbala, J. (2013)*: Fluid mechanics fundamentals and applications (SI units)*. McGraw Hill.